Adding High-Content Screening annotations to 3DBionotes-WS

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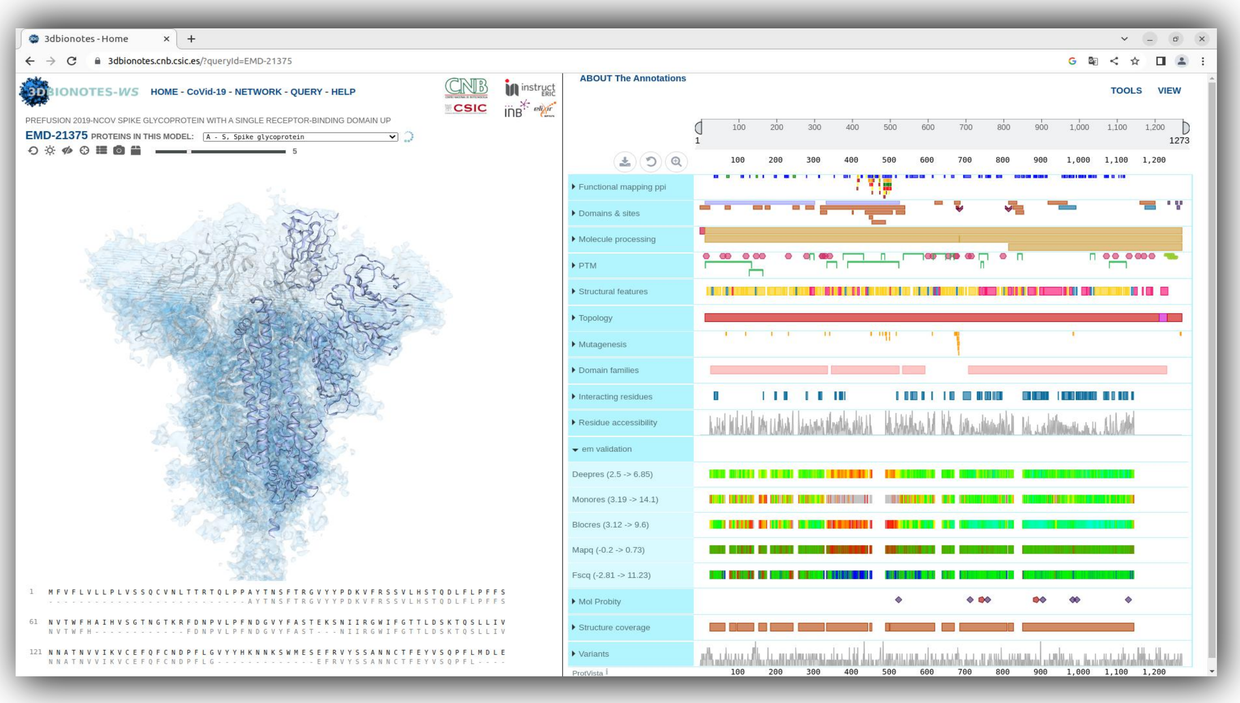
What is 3DBionotes-WS

3DBionotes-WS is a **web-based 3D viewer** for macromolecular complexes that allows you to display the different **atomic models** and their related **cryo-EM density maps** deposited at public repositories such as the Protein Data Bank (PDB) and the Electron Microscopy Data Bank (EMDB).

The **added value** of 3DBionotes-WS is that allows the user to interact with the molecular structures and provides a bunch of **additional annotations** that give a better understanding of the **biological context** and **features** of those complexes.



Motivation

3DBionotes-WS is always in continuous expansion and looking for new data types to incorporate, aiming to enhance **access** and **reuse** of **FAIR research outputs.** Currently, we are focused on those coming out of image-based experiments.

Mission

Create a **connection between high-level image data** and the **structural information** behind it to extend the context of macromolecular complexes, concentrating first on imaging data from **High-Content Screening** (HCS) assays as a development example, but being as general as possible in the future.

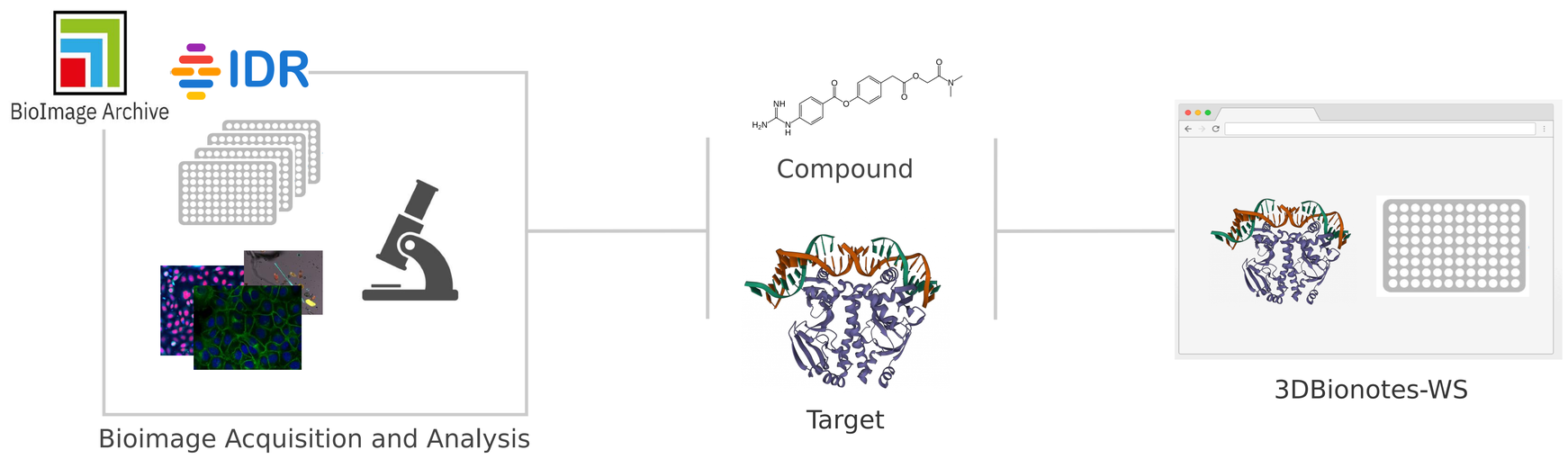


Image data is collected from public repositories such as the Image Data Resource (IDR) and the BioImage Archive (BIA), which is then linked to a specific compound and target and displayed in the interactive 3D environment of 3DBionotes-WS.

Contributing to EOSC

We are part of the **Test Science Project** "Open Imaging Data Sharing in EOSC / COVID-19 as Demonstrator" within **EOSC Future.** Here we work together with other **partners** within the imaging field and **EOSC Core/Horizontal Services** providers to enable you access to the high-throughput image, structural, and chemical screening datasets from around the globe.

You can find us in the EOSC Marketplace!
https://marketplace.eosc-portal.eu/services/3dbionotes-ws?q=3DBionotes-WS

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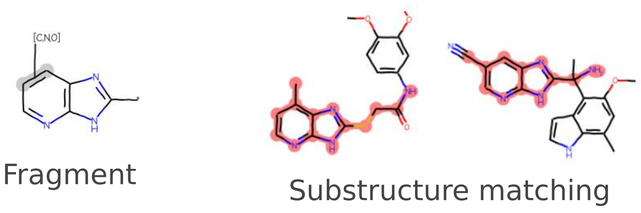
Use Case

3DBionotes-WS users will be able to search for **COVID-19-related macromolecular structures** bound to a collection of compounds with **potential clinical use**, such as Camostat, and get a summary of the possible therapeutic strategies.

Future Plans

We are working on a **substructure matching tool** based on RDKit to provide a set of **small fragments** that there is evidence that binds specific proteins and that are also part of well-defined compounds and **potential drug candidates** proven in HCS assays.



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Learn more about 3DBionotes-WS
https://3dbionotes.cnb.csic.es/ws

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